AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 11 · STUDENT LESSON

Visual Composition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 5: separate multiple constraints in a prompt.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of image generation and provenance. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of image generation and **provenance**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

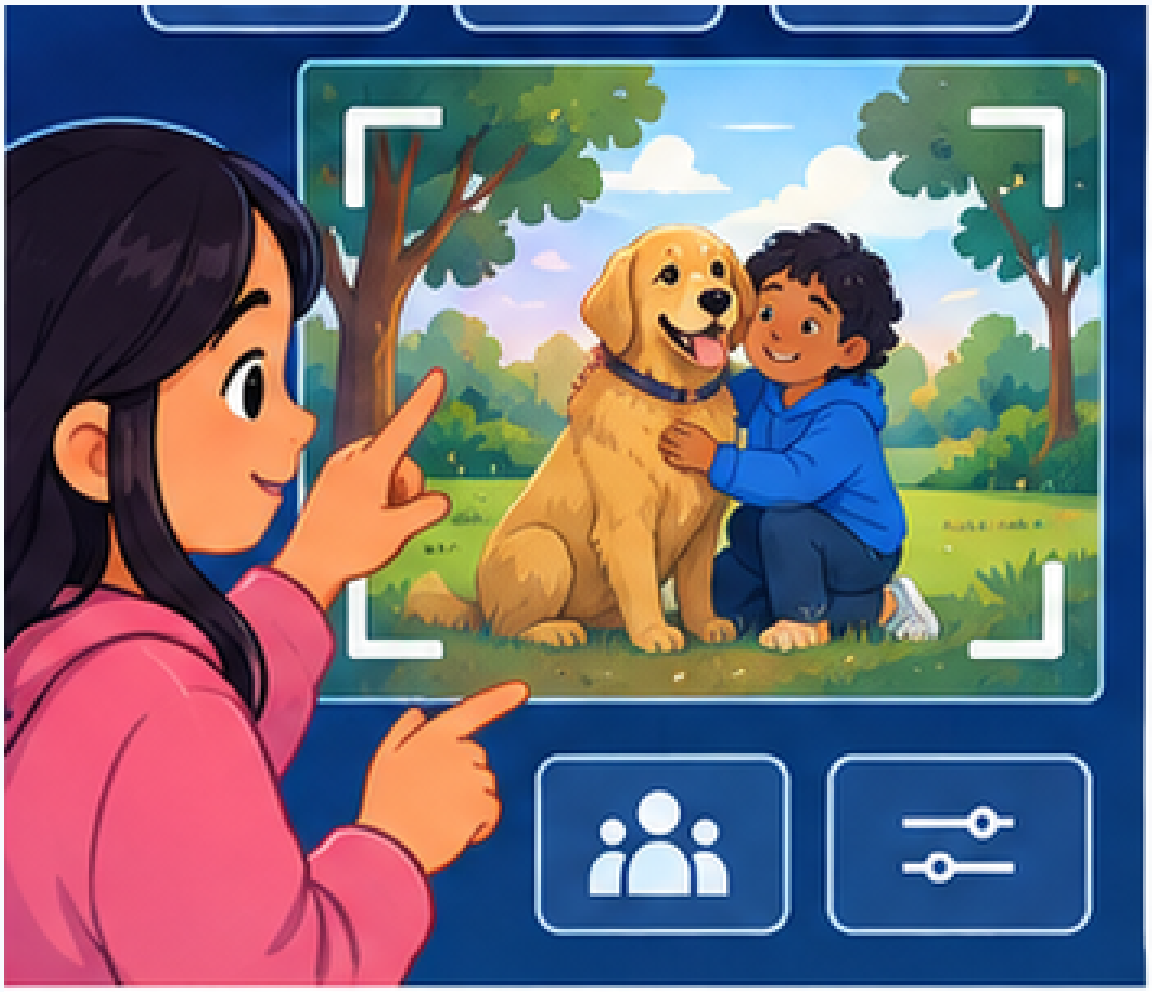
Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 11: Visual Composition

Subject, **setting**, **framing**, **relationship**s, and constraints shape an image request.



CREATOR MISSION

Diagnose and repair three ambiguous visual prompts.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Generation failures and verification. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can check object counts and spatial relations as separate visual requirements.

STUDENT LESSON · 2 OF 18

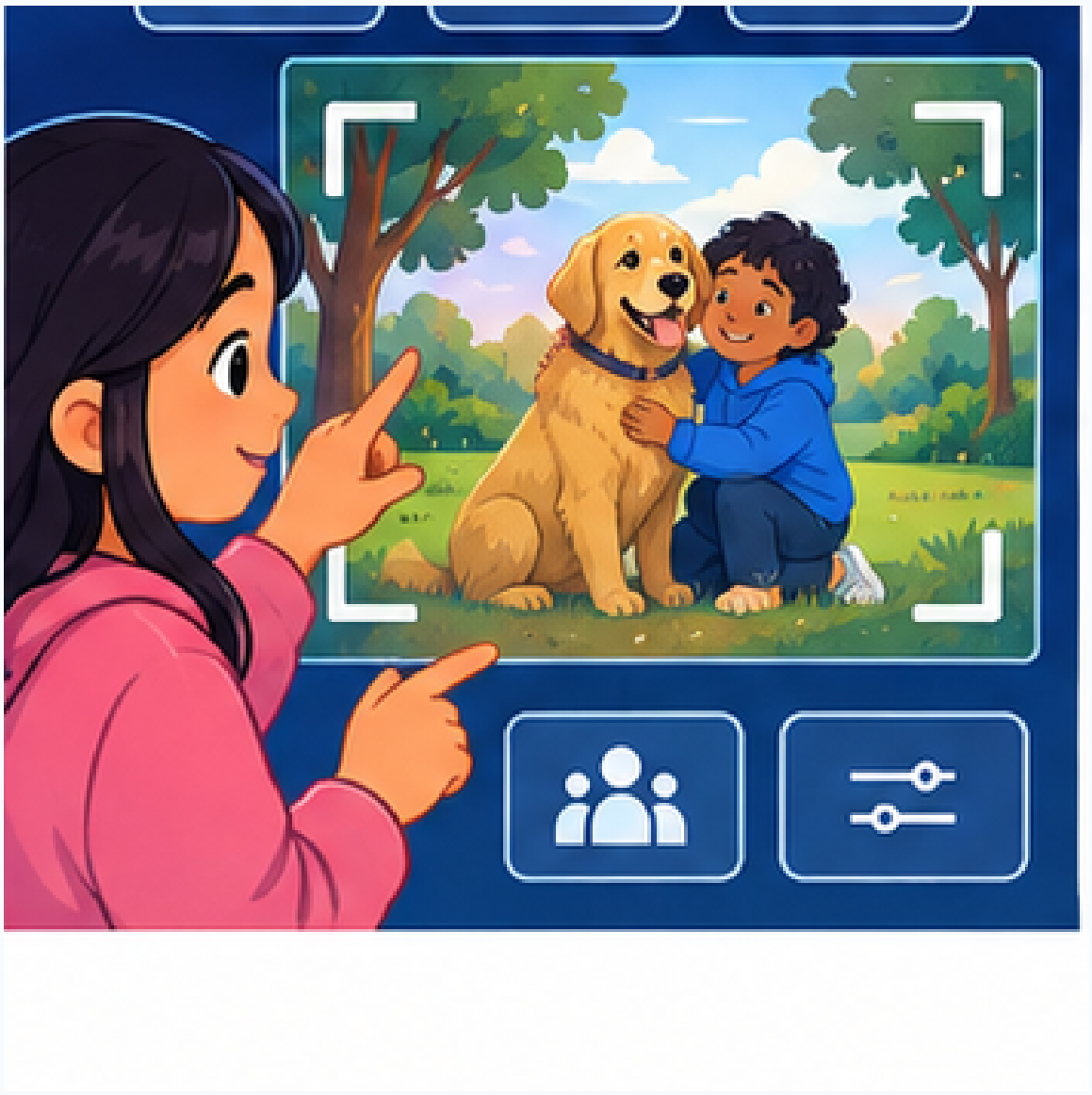
Notice the evidence

> Investigate the supplied Visual Composition case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use visual composition to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

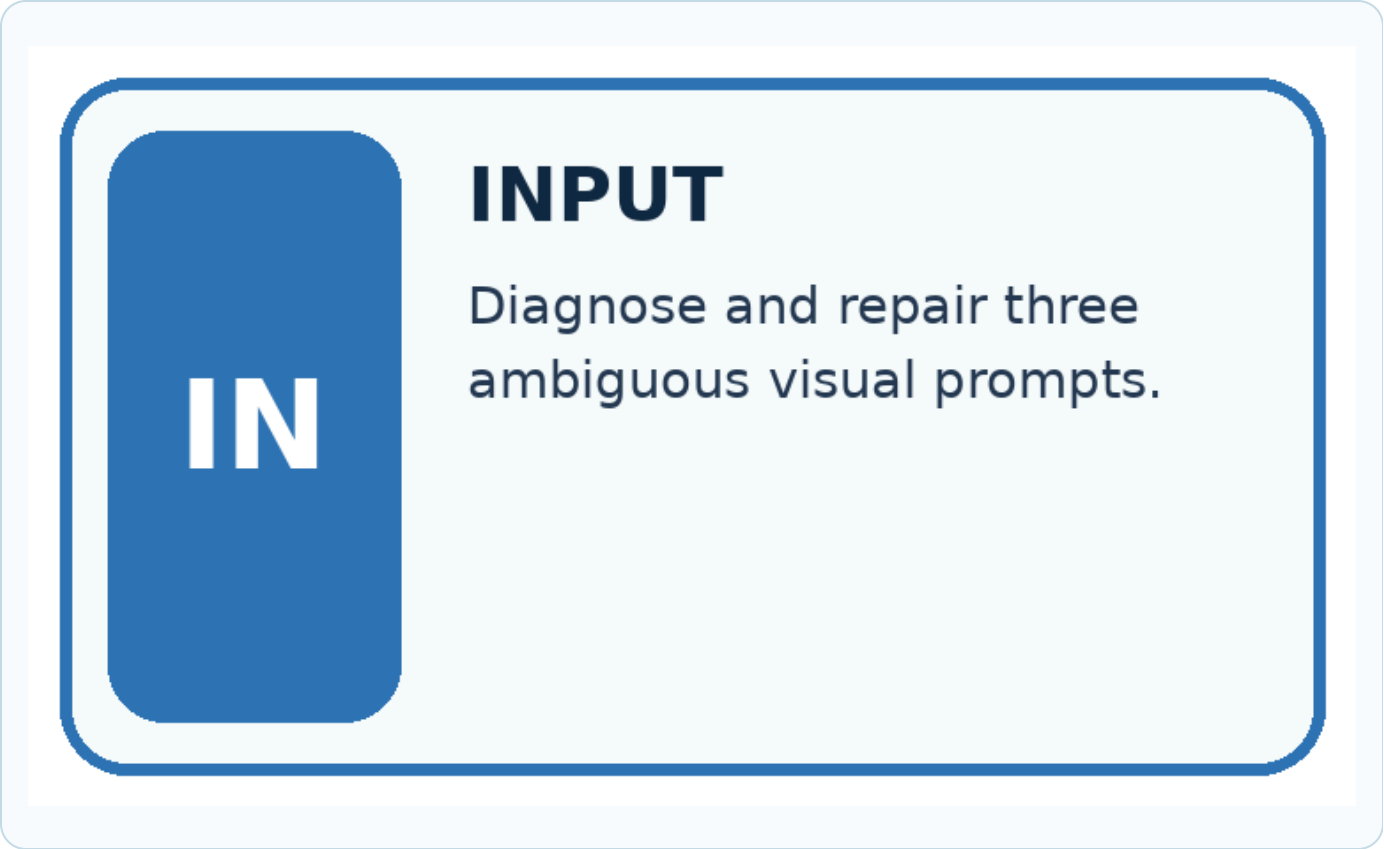
See how it works

> Follow the mechanism for Visual Composition. Connect each labelled input, deciding step and result.

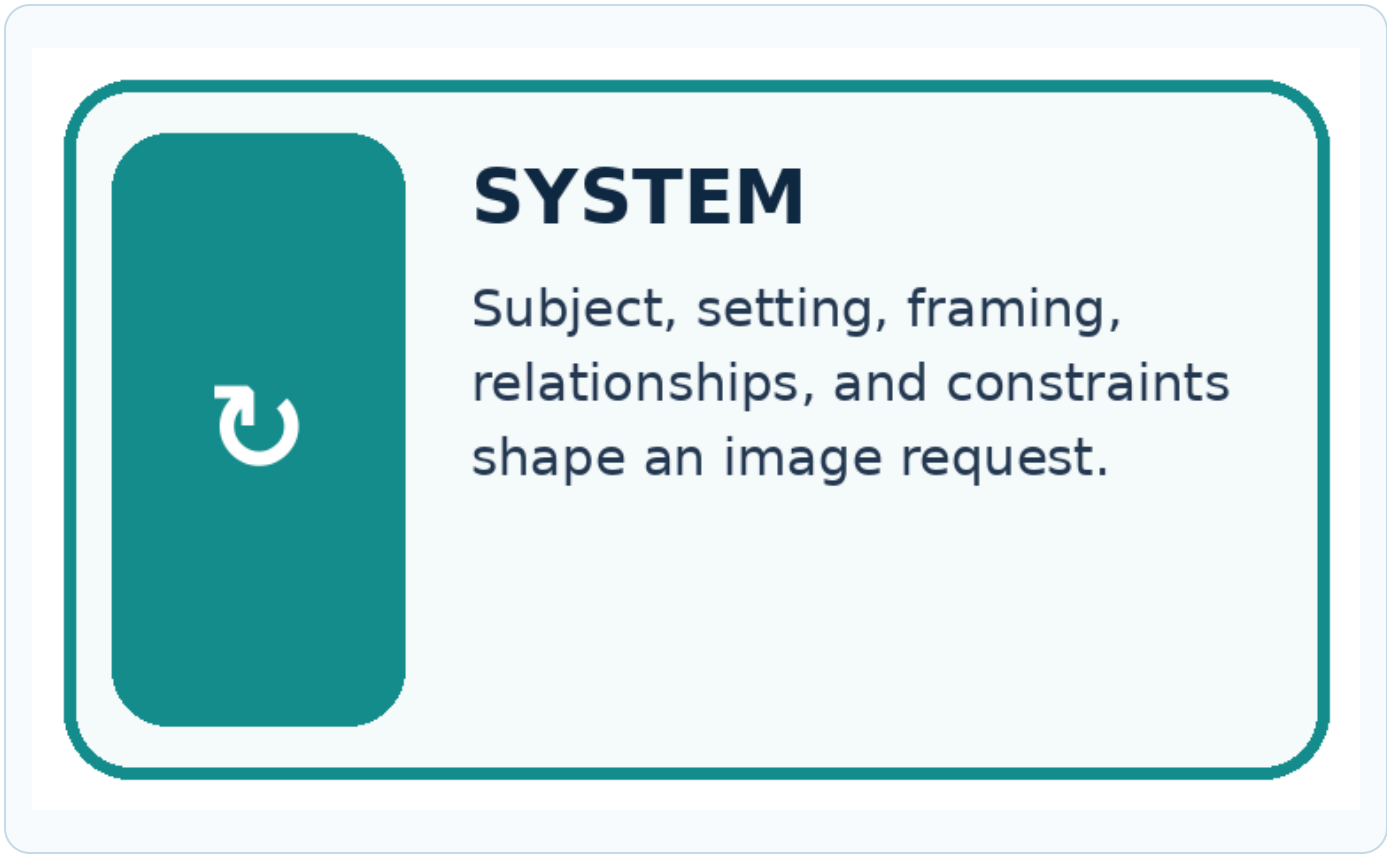
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Subject, setting, framing, relationships, and constraints shape an image request.
↓
OUT
EVIDENCE
Diagnose and repair three ambiguous visual prompts.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Image generation and provenance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.	Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.	It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

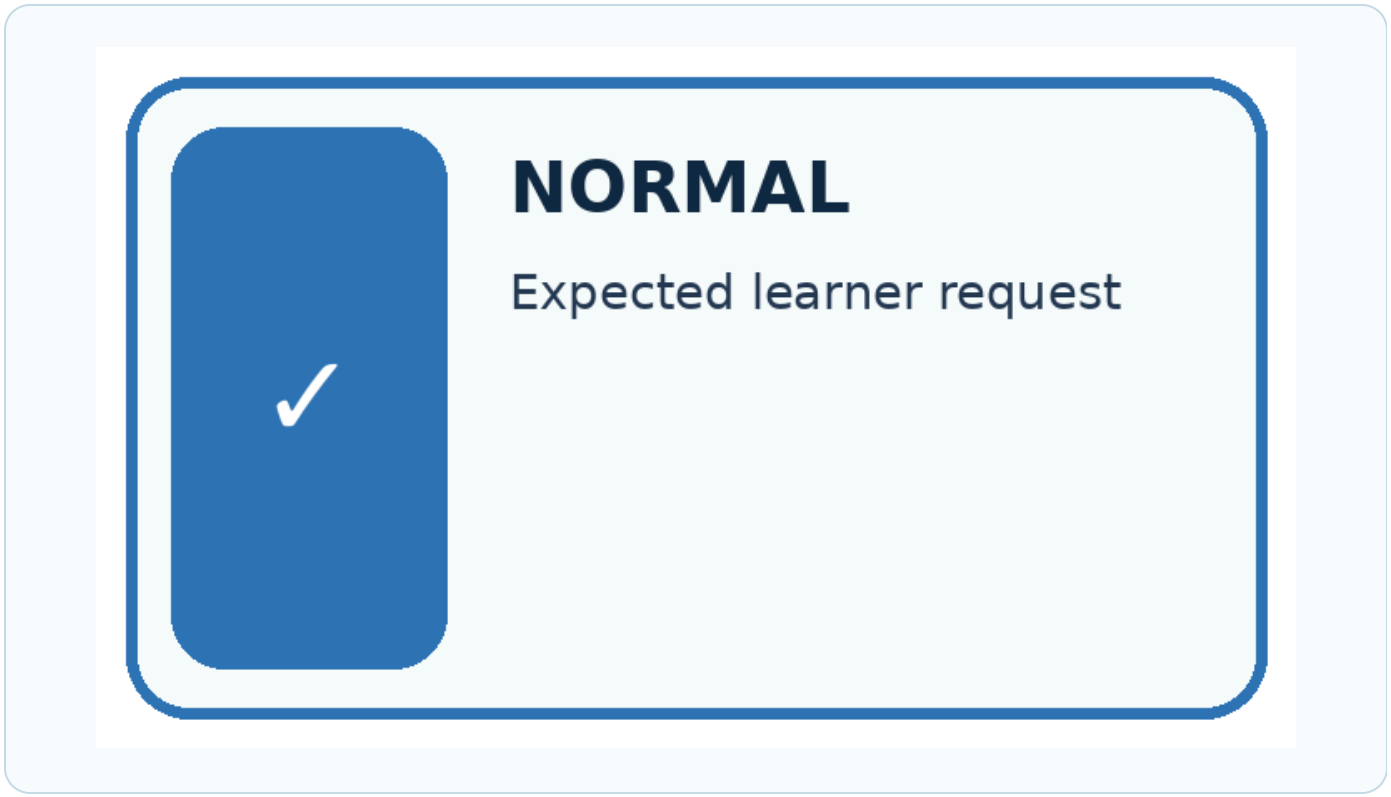
> Compare the supplied cases for Visual Composition. Identify the detail that changes the conclusion.

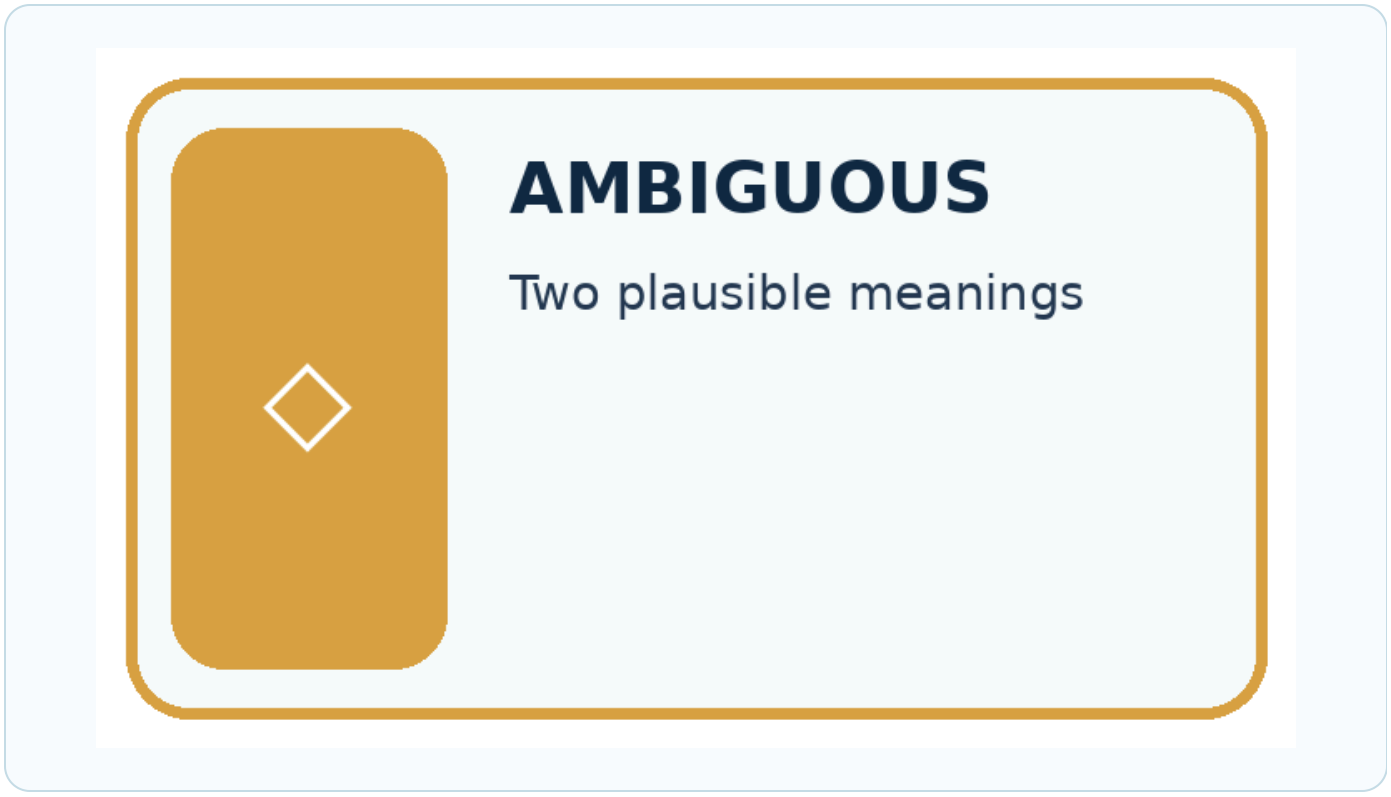
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Predicted result

Prediction: Position now passes; arrow direction still fails.

Why the result follows

Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

STUDENT LESSON · 5 OF 18

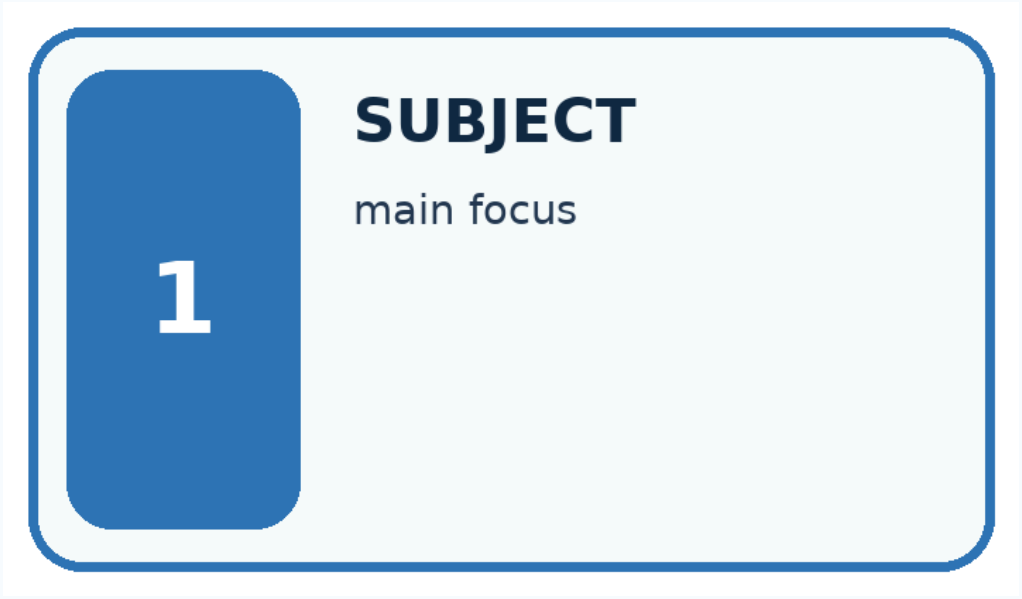
Words that unlock the idea

> Use the vocabulary of Visual Composition to explain a concrete decision, rather than only memorising definitions.

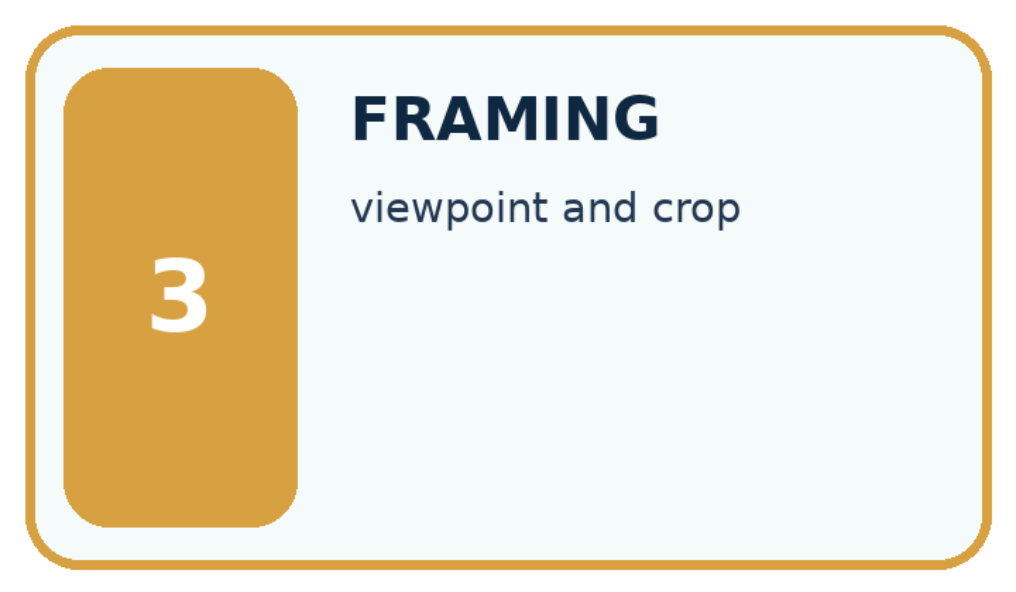
VOCABULARY LAB

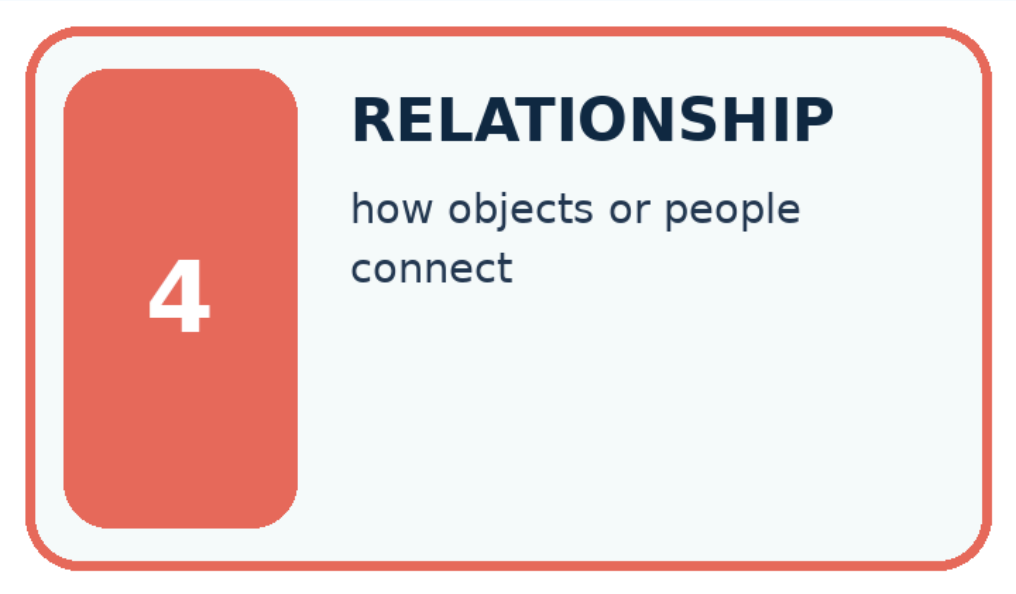
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
SUBJECT
main focus


2
SETTING
environment around the subject





Word	My own example	Not this / boundary
subject	The main person, object or topic represented in an image or piece of content.	
setting	A chosen configuration value that affects how a system operates.	

Word	My own example	Not this / boundary
framing	The way a question or situation is presented, which can influence interpretation.	
relationship	A connection between quantities, events or concepts; an association need not be causal.	

Words used in the closer explanation

Word	Meaning
Provenance	Recorded origin and transformation history
Visual criterion	A stated requirement checked against an image
Authenticity claim	A claim about the source or reality of depicted content

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Visual Composition, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 11 • WORKED EVIDENCE

1. Prompt names a robot and child but not their relationship.
2. Model may place them far apart or reverse who teaches whom.
3. Add action, viewpoint, spacing, and must-show constraints.
4. Evaluate whether composition communicates the learning idea without misleading detail.

1. Prompt names a robot and child but not their **relationship**.

2. Model may place them far apart or reverse who teaches whom.

3. Add action, viewpoint, spacing, and must-show constraints.

4. Evaluate whether composition communicates the learning idea without misleading detail.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Worked explanation · Visual Composition

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Visual Composition

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

The facts we will use

Supplied facts

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied facts

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

AI ACADEMY / AI-3/11

Visual Composition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count objects

Two bottles and one beaker satisfy the count requirement.



2. Check position

Bottles on the right fail the required left-of relation.



3. Check direction

Arrow from beaker to bottles reverses the required flow.



4. Repair brief

Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

CHANGE ONE THING

Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count objects	Two bottles and one beaker satisfy the count requirement.
2. Check position	Bottles on the right fail the required left-of relation.
3. Check direction	Arrow from beaker to bottles reverses the required flow.
4. Repair brief	Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

What this example does not establish

Keep the stated limit

A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

Keep the stated limit

Read the labelled picture: Which checks pass after that single change, and what still needs repair?

Keep the stated limit

Change one thing: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Visual Composition claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More visual adjectives always create a clearer composition. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

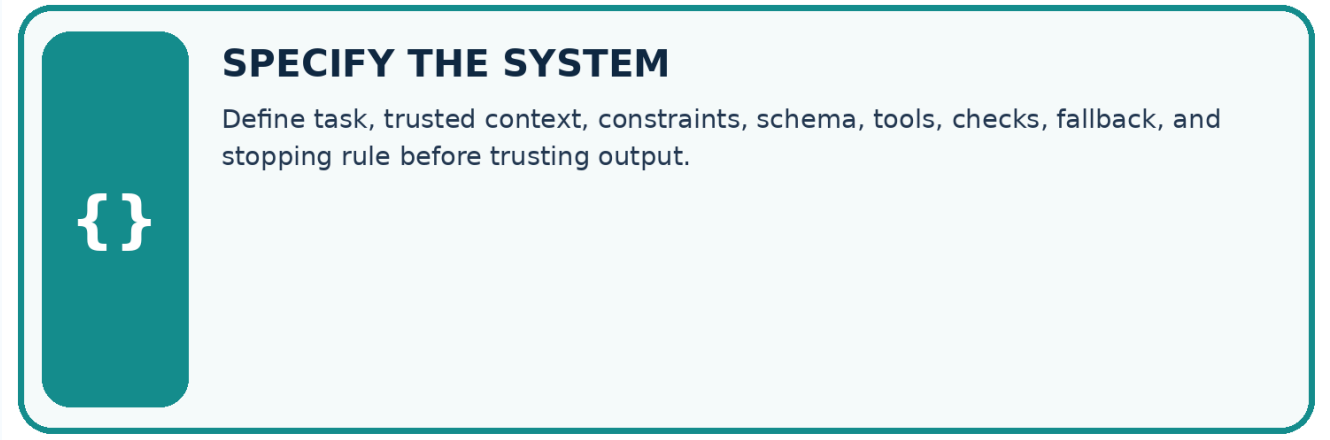
Practise together

> Practise Visual Composition with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and **provenance**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which checks pass after that single change, and what still needs repair?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Visual Composition. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Repair three prompts by changing one missing composition element; sketch expected **framing** before generation and compare.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

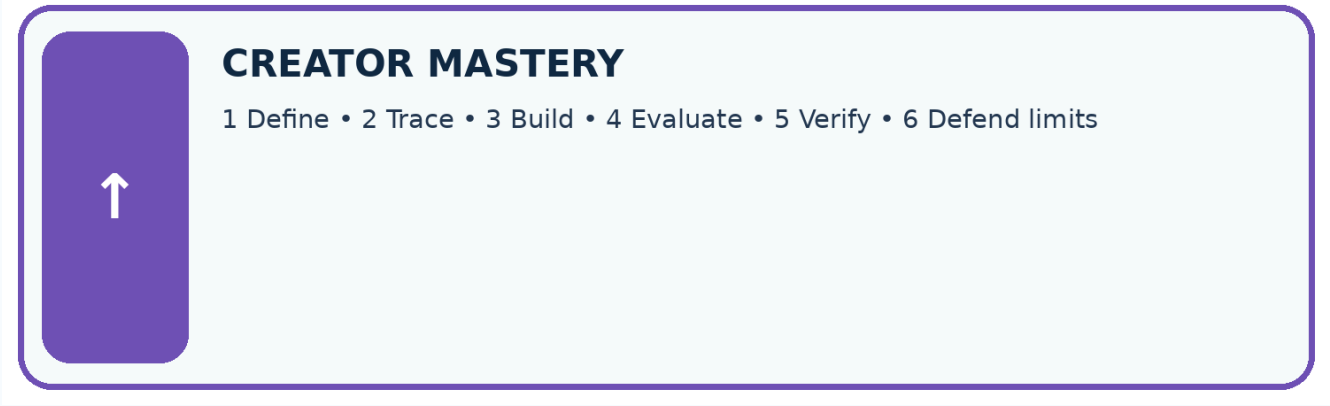
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Visual Composition. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

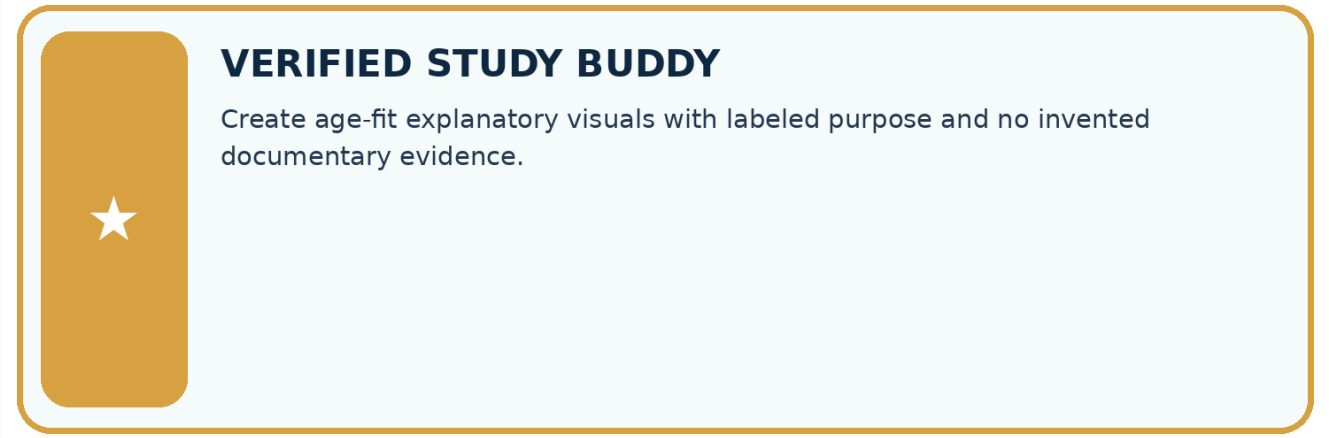
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

STUDENT LESSON · 12 OF 18

Show what you know

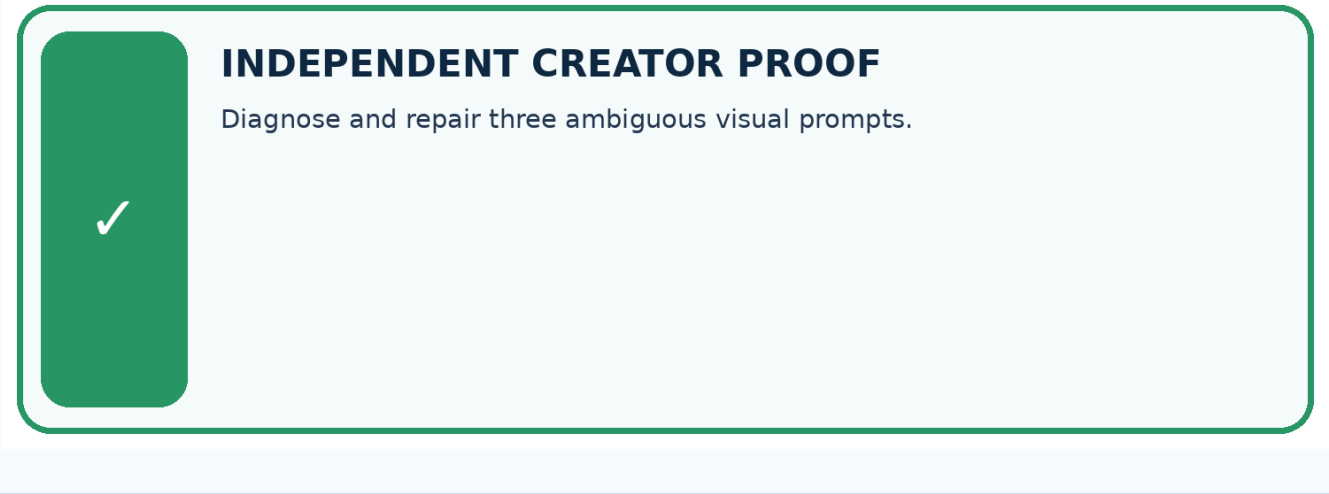
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Diagnose and repair three ambiguous visual prompts.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Visual Composition case for supported practice. Repeated examples do not become fresh assessments.

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

The question

Create a visual checklist.

Worked explanation

Check **subject**s, relative size, positions and sky separately. The position violates composition even if **subject**s exist.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Map the brief to regions

Evidence or stage	Value or result
Upper region	Open sky
Left foreground	Large tree
Right foreground	Small bicycle

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

Create a visual checklist.

Check **subject**s, relative size, positions and sky separately. The position violates composition even if **subject**s exist.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle.

Evaluate this description.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Visual Composition, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

By the end, I can	How I will show it
Explain the mechanism and terminology of image generation and provenance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Visual Composition without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of image generation and **provenance**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

1. Check position

2. Repair brief

3. Count objects

4. Check direction

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage

that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Position now passes; arrow direction still fails.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair a missing building block

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and **provenance**.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

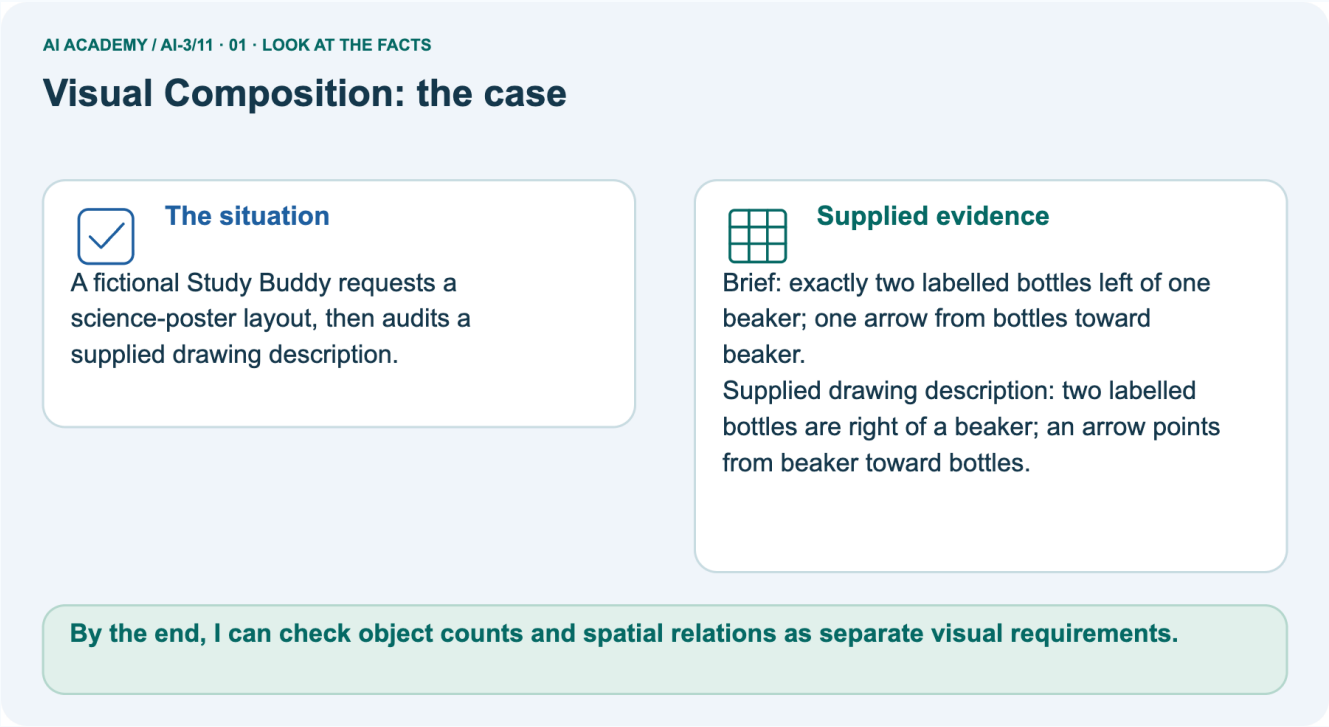
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Visual Composition graphic using its labels, deciding evidence and relevant limitation.

01. Visual Composition: the case



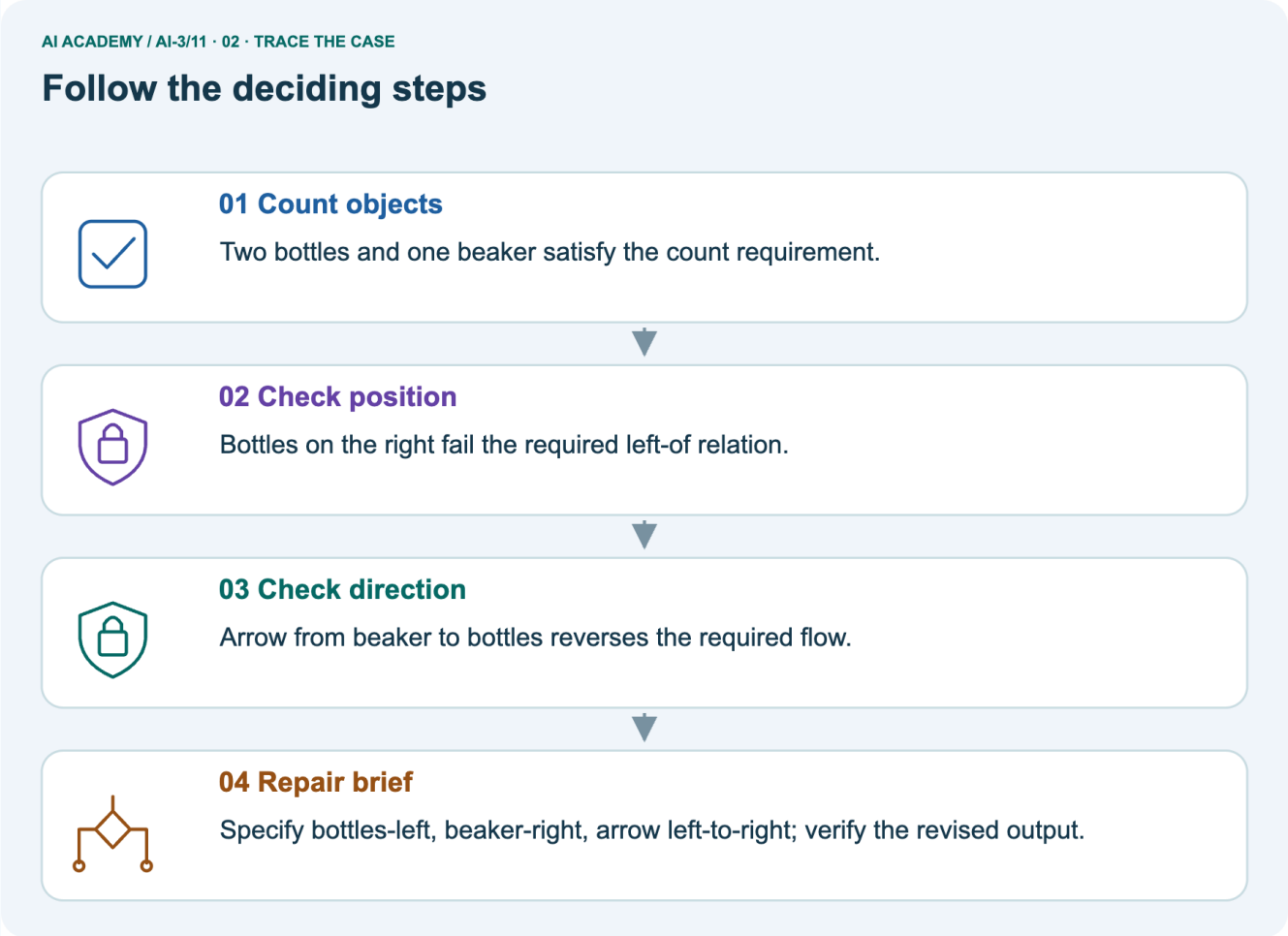
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description. Supplied facts: Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

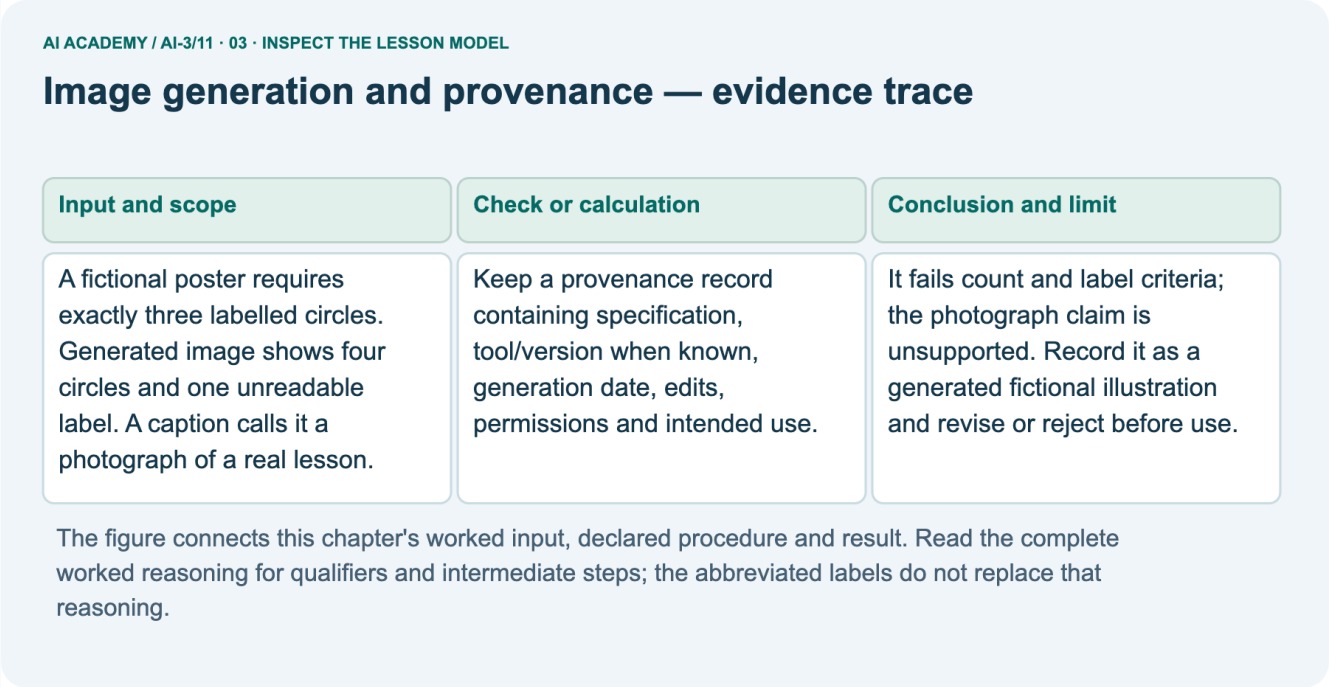


Follow the labelled picture with the matching explanation. Open the vector diagram

Count objects: Two bottles and one beaker satisfy the count requirement. → Check position: Bottles on the right fail the required left-of relation. → Check direction: Arrow from beaker to bottles reverses the required flow. → Repair brief: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Image generation and provenance — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

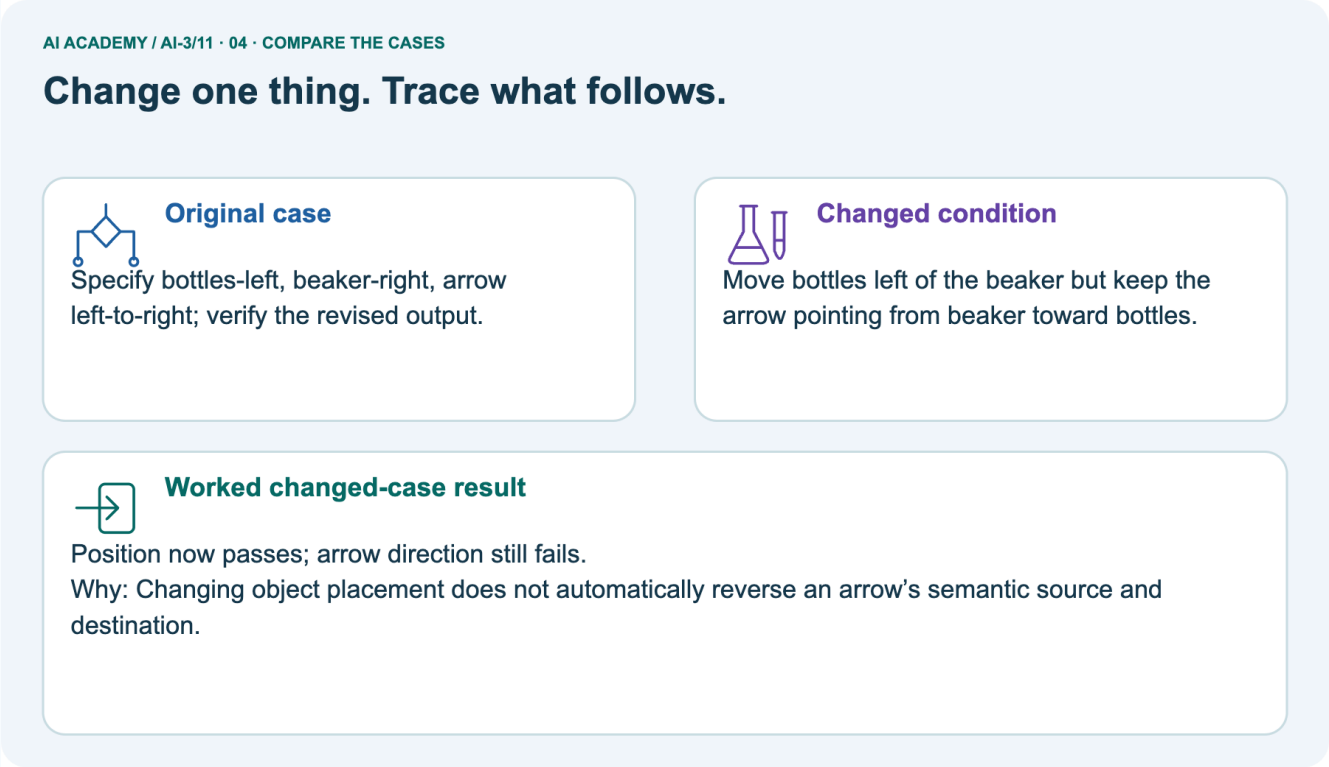
Image generation and **provenance** — **evidence** trace. A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label.

A caption calls it a photograph of a real lesson. | Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. | It fails count and label criteria; the photograph claim is unsupported.

Record it as a generated fictional illustration and revise or reject before use.

Read the labels and values. Explain the **relationship** the model shows.

04. Change one thing. Trace what follows.



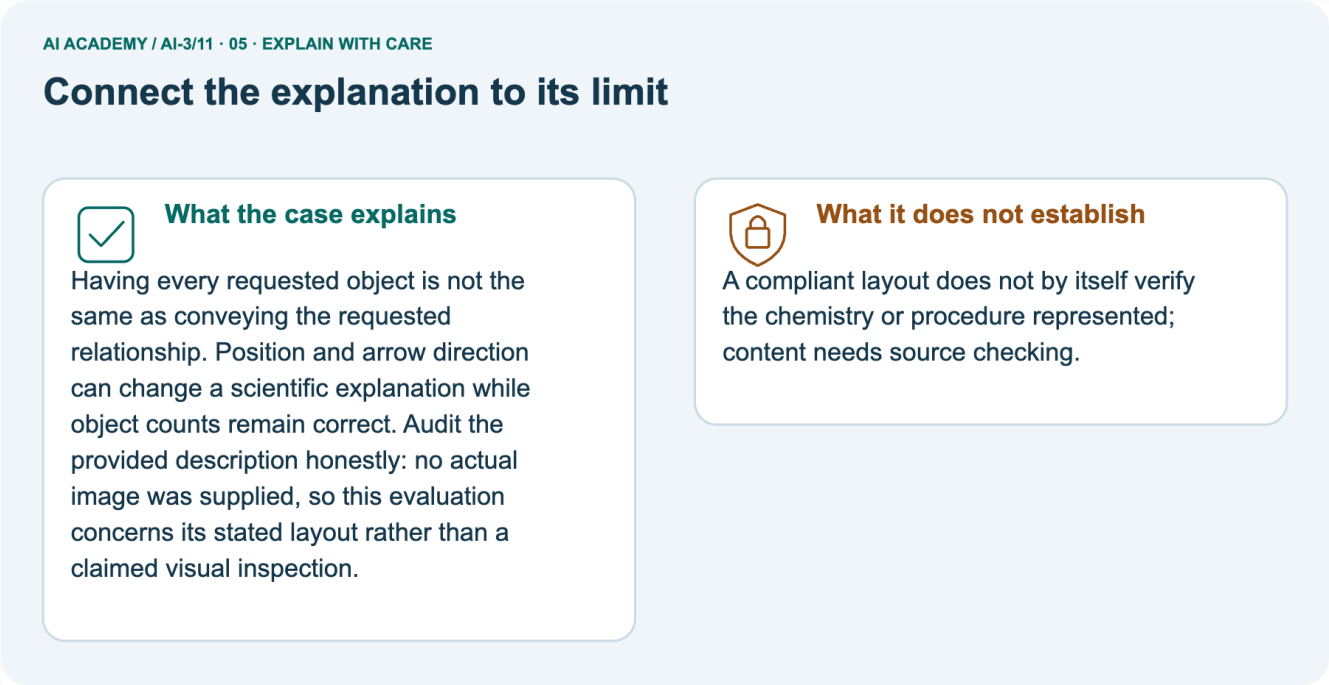
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.
Changed condition: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Expected result: Position now passes; arrow direction still fails. Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

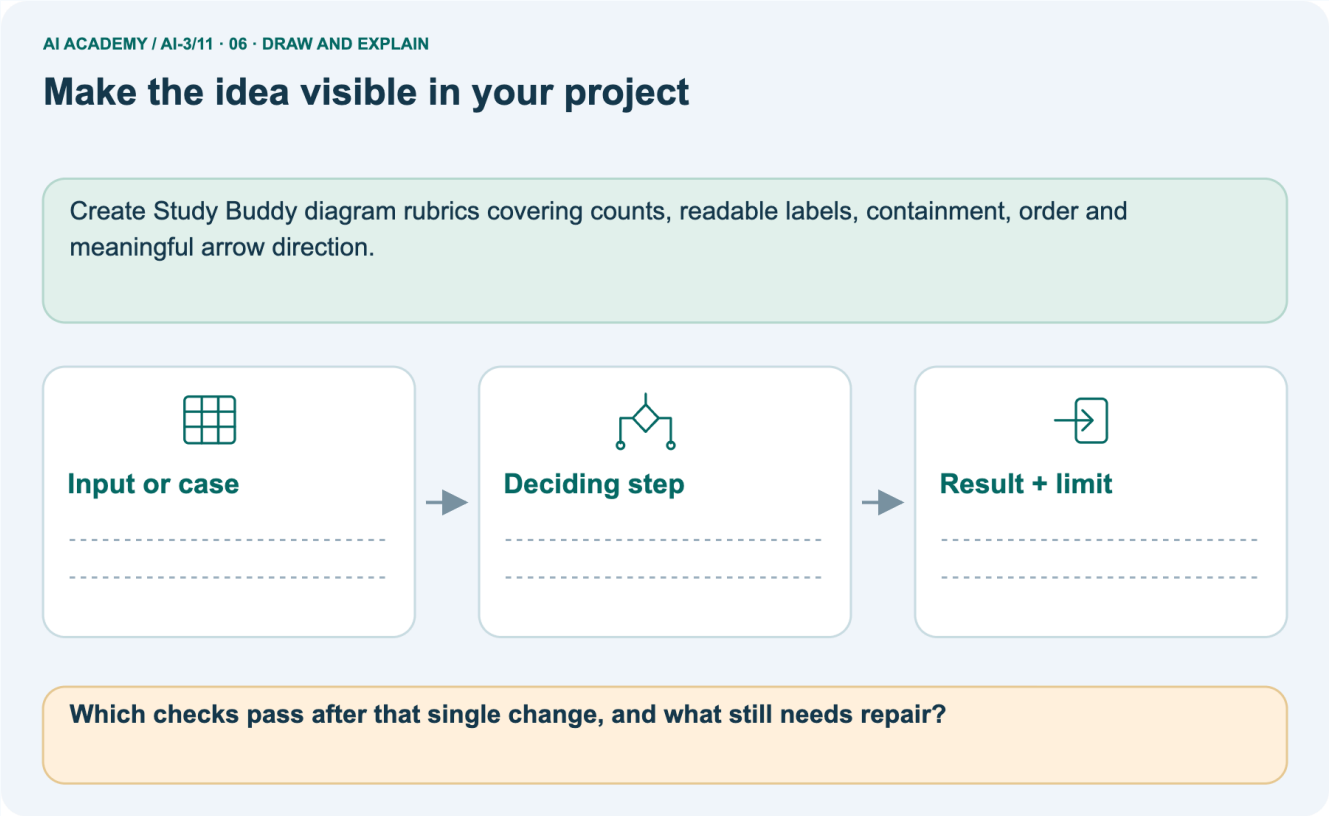
Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Limit: A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction. Draw your **input**, deciding step, result and limit.

Supported question: Which checks pass after that single change, and what still needs repair?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.